Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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1. Цели и задачи работы

Цель лабораторной работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

Задачи лабораторной работы

1 Изучить перенаправление ввода-вывода

2 Изучить работу фильтров

3 Изучить команду поиска

4 Ознакомиться с управлением процессами

5 Ознакомиться с командами df du

2. Процесс выполнения лабораторной работы

Перенаправление ввода-вывода

```
ysinavbarova@ysinavbarova:~$ ls /etc/ > file.txt
ysinavbarova@ysinavbarova:~$ ls >> file.txt
ysinavbarova@ysinavbarova:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
```

Рисунок 1: Запись в файл

Работа фильтра

```
ysinavbarova@ysinavbarova:~$ grep .conf file.txt > conf.txt
ysinavbarova@ysinavbarova:~$
cat conf.txt
ysinavbarova@ysinavbarova:~$ cat conf.txt
anthy-unicode.conf
asound.conf
brltty.conf
chrony.conf
cpupower-service.conf
dconf
dnsmasq.conf
dracut.conf
dracut.conf.d
fprintd.conf
fuse.conf
host.conf
idmapd.conf
kdump.conf
krb5.conf
krb5.conf.d
ld.so.conf
ld.so.conf.d
libaudit.conf
libuser.conf
locale.conf
logrotate.conf
```

Рисунок 2: Поиск расширения .conf

Команда поиска

```
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage03/report/_resources/csl
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage03/report/bib/cite.bib
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage04/report/_resources/csl
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage04/report/bib/cite.bib
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage05/report/_resources/csl
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage05/report/bib/cite.bib
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage06/report/_resources/csl
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/project-personal/stage06/report/bib/cite.bib
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/presentation/report/_resources/csl
/home/ysinavbarova/work/study/2025-2026/Операционные системы/2026-1--study--os-i
ntro/presentation/report/bib/cite.bib
/home/ysinavbarova/git-extended/.git/hooks/commit-msg.sample
/home/ysinavbarova/git-extended/.git/objects/c3
/home/ysinavbarova/git-extended/.git/config
/home/ysinavbarova/conf.txt
ysinavbarova@ysinavbarova:~$
```

Рисунок 3: Поиск файлов

Команда поиска

```
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: ‘/etc/lvm/archive’: Permission denied
/etc/logrotate.d/httpd
find: ‘/etc/lvm/backup’: Permission denied
find: ‘/etc/lvm/cache’: Permission denied
find: ‘/etc/lvm/devices’: Permission denied
find: ‘/etc/nftables’: Permission denied
find: ‘/etc/openvpn/client’: Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: ‘/etc/openvpn/server’: Permission denied
find: ‘/etc/pki/rsyslog’: Permission denied
find: ‘/etc/polkit-1/localauthority’: Permission denied
find: ‘/etc/polkit-1/rules.d’: Permission denied
find: ‘/etc/sos/cleaner’: Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
:find: ‘/etc/ssh/sshd_config.d’: Permission denied
find: ‘/etc/sssd’: Permission denied
find: ‘/etc/sudoers.d’: Permission denied
find: ‘/etc/userdb’: Permission denied
```

Рисунок 4: Поиск файлов

Управление процессами

```
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ find /etc -name "h*" -print | less
find: '/etc/audit': Permission denied
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ find ~ -name "log*" > logfile &
[1] 6559
ysinavbarova@ysinavbarova:~$
[1]+  Done                    find ~ -name "log*" > logfile
ysinavbarova@ysinavbarova:~$ rm logfile
ysinavbarova@ysinavbarova:~$
```

Рисунок 5: Фоновый запуск процесса

Управление процессами

```
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ find /etc -name "h*" -print | less
find: ‘/etc/audit’: Permission denied
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ find ~ -name "log*" > logfile &
[1] 6559
ysinavbarova@ysinavbarova:~$
[1]+  Done                    find ~ -name "log*" > logfile
ysinavbarova@ysinavbarova:~$ rm logfile
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ gedit &
[1] 6597
ysinavbarova@ysinavbarova:~$ ps | grep gedit
   6597 pts/1    00:00:00 gedit
ysinavbarova@ysinavbarova:~$ kill 6597
[1]+  Terminated              gedit
ysinavbarova@ysinavbarova:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

Команды df и du

```
DF(1)                          User Commands                          DF(1)

NAME
       df - report file system space usage

SYNOPSIS
       df [OPTION]... [FILE]...

DESCRIPTION
       This  manual  page  documents  the  GNU  version of df.  df displays the
       amount of space available on the file system containing each  file  name
       argument.   If  no  file  name is given, the space available on all cur-
       rently mounted file systems is shown.  Space is shown in  1K  blocks  by
       default,  unless  the  environment  variable  POSIXLY_CORRECT is set, in
       which case 512-byte blocks are used.

       If an argument is the absolute file name of a device node  containing  a
       mounted  file  system,  df shows the space available on that file system
       rather than on the file system containing the device node.  This version
       of df cannot show the space available on unmounted file systems, because
       on most kinds of systems doing so requires non-portable intimate  knowl-
       edge of file system structures.

OPTIONS
       Show  information  about  the file system on which each FILE resides, or
       all file systems by default.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              include pseudo, duplicate, inaccessible file systems

 Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

Команды df и du

```
DU(1)                          User Commands                          DU(1)

NAME
       du - estimate file space usage

SYNOPSIS
       du [OPTION]... [FILE]...
       du [OPTION]... --files0-from=F

DESCRIPTION
       Summarize device usage of the set of FILEs, recursively for directories.

       Mandatory arguments to long options are mandatory for short options too.

       -0, --null
              end each output line with NUL, not newline

       -a, --all
              write counts for all files, not just directories

       --apparent-size
              print  apparent  sizes rather than device usage; although the ap-
              parent size is usually smaller, it may be larger due to holes  in
              ('sparse')  files,  internal  fragmentation, indirect blocks, and
              the like

       -B, --block-size=SIZE
              scale sizes by SIZE before  printing  them;  e.g.,  '-BM'  prints
              sizes in units of 1,048,576 bytes; see SIZE format below

       -b, --bytes
              equivalent to '--apparent-size --block-size=1'
 Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

Команды df и du

```
ysinavbarova@ysinavbarova:~$
ysinavbarova@ysinavbarova:~$ df
Filesystem     1K-blocks     Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 91248168  63340776  60% /
devtmpfs         4011284        0   4011284   0% /dev
tmpfs            4035656       92   4035564   1% /dev/shm
tmpfs            1614264     1944   1612320   1% /run
tmpfs               1024        0      1024   0% /run/credentials/systemd-journa
ld.service
tmpfs            4035660       80   4035580   1% /tmp
/dev/nvme0n1p3 155186176 91248168  63340776  60% /home
/dev/nvme0n1p2   1992552   518800   1352512  28% /boot
/dev/loop2         49280    49280         0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0         68480    68480         0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1        115712   115712         0 100% /var/lib/snapd/snap/hugo/25676
tmpfs               1024        0      1024   0% /run/credentials/systemd-resolv
ed.service
tmpfs             807128      168    806960   1% /run/user/1062
/dev/sr0         2677920  2677920         0 100% /run/media/ysinavbarova/Fedora-
WS-Live-43
ysinavbarova@ysinavbarova:~$
```

Рисунок 9: Справка по команде du

Команды df и du

```
4       ./git-extended/.git/objects/9f
4       ./git-extended/.git/objects/88
4       ./git-extended/.git/objects/2b
4       ./git-extended/.git/objects/c3
4       ./git-extended/.git/objects/89
4       ./git-extended/.git/objects/2e
4       ./git-extended/.git/objects/a3
76      ./git-extended/.git/objects
8       ./git-extended/.git/logs/refs/heads
8       ./git-extended/.git/logs/refs/remotes/origin
8       ./git-extended/.git/logs/refs/remotes
16      ./git-extended/.git/logs/refs
20      ./git-extended/.git/logs
212     ./git-extended/.git
220     ./git-extended
0       ./monthly
0       ./reports/monthly/monthly
0       ./reports/monthly
0       ./reports
4       ./ski.plases/equipment
0       ./ski.plases/plans
4       ./ski.plases
0       ./australia
0       ./play/games/play
0       ./play/games
0       ./play
823252  .
ysinavbarova@ysinavbarova:~$
```

Рисунок 10: Запуск команды du

Команда поиска

```
/home/ysinavbarova/git-extended/.git/objects/d5
/home/ysinavbarova/git-extended/.git/objects/1b
/home/ysinavbarova/git-extended/.git/objects/9f
/home/ysinavbarova/git-extended/.git/objects/88
/home/ysinavbarova/git-extended/.git/objects/2b
/home/ysinavbarova/git-extended/.git/objects/c3
/home/ysinavbarova/git-extended/.git/objects/89
/home/ysinavbarova/git-extended/.git/objects/2e
/home/ysinavbarova/git-extended/.git/objects/a3
/home/ysinavbarova/git-extended/.git/logs
/home/ysinavbarova/git-extended/.git/logs/refs
/home/ysinavbarova/git-extended/.git/logs/refs/heads
/home/ysinavbarova/git-extended/.git/logs/refs/remotes
/home/ysinavbarova/git-extended/.git/logs/refs/remotes/origin
/home/ysinavbarova/monthly
/home/ysinavbarova/reports
/home/ysinavbarova/reports/monthly
/home/ysinavbarova/reports/monthly/monthly
/home/ysinavbarova/ski.plases
/home/ysinavbarova/ski.plases/equipment
/home/ysinavbarova/ski.plases/plans
/home/ysinavbarova/australia
/home/ysinavbarova/play
/home/ysinavbarova/play/games
/home/ysinavbarova/play/games/play
ysinavbarova@ysinavbarova:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

Вывод

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.